

Newton's Laws of Motion - teacher-authored staging test source.

Newton's first law: an object remains at rest or at constant velocity unless a nonzero resultant external force acts on it. Mass measures inertia.

Newton's second law: for constant mass in an inertial frame, the resultant external force is $F = ma$. Force is measured in newtons. A 2 kg object accelerating at 3 m/s^2 has resultant force 6 N.

Newton's third law: interacting bodies exert equal and opposite forces on each other. These forces act on different bodies and do not cancel on one object's free-body diagram.

A free-body diagram shows external forces on a chosen object. Choose the object, identify its forces, find the resultant, and use $F = ma$ to solve force and acceleration problems.

These teacher-authored notes are not historical examination questions.